Microsoft 365 ↔ Google Workspace Coexistence (Mail Routing, Free/Busy, Directory Sync)

Implementation Solution Guide

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Executive Summary — Microsoft 365 ↔ Google Workspace Coexistence

The situation

For the months it takes to migrate between **Microsoft 365** and **Google Workspace**, both platforms run side by side. Users don't care which platform holds their colleague's mailbox — they expect mail to arrive, calendars to show availability, and the directory to list everyone. Without deliberate coexistence, the split organization breaks in visible, painful ways: mail bounces (NDRs), meeting scheduling shows blank calendars, and people can't find each other in the address book.

What this solution does

It designs and implements **coexistence** — three interop channels that make Microsoft and Google behave as **one organization** until the migration completes:

1. **Mail routing / split delivery** — the shared domain is set to relay: each platform delivers to its own mailboxes and forwards mail for everyone else to the other platform. No NDRs during the split.
1. **Free/busy calendar interop** — Google Calendar Interop plus Exchange availability sharing let Outlook and Google Calendar resolve each other's availability **in both directions**.
1. **Directory sync** — one identity source projects users and cross-platform contacts into both ecosystems (Google Cloud Directory Sync + Exchange mail contacts), producing a single unified address book.

How it's delivered

The engagement follows a disciplined **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** model with explicit pass/fail gates and a documented rollback at every step. Every change — DNS, mail connectors, calendar endpoints, directory rules — is proven on a small pilot before it touches the fleet, because mail-flow and DNS changes are high-impact. Five supporting scripts automate readiness discovery, per-user interop validation, and coverage reporting; the two that make changes default to a safe preview (`-WhatIf`) mode.

The outcome

Microsoft and Google working as one through the transition — mail routes, calendars resolve, the directory stays in sync — piloted, validated, and reversible. Because coexistence is a *transition state*, the solution also builds in a **clean decommission**: once consolidation completes, the legacy platform is retired in a controlled sequence with nothing left behind.

Why it matters

Coexistence is the difference between a migration users barely notice and one that generates a flood of "my email bounced" and "I can't see anyone's calendar" tickets for months. This solution turns the riskiest

window of a platform migration into a **quiet, unified experience** — self-guided, tested, and at a fraction of the cost of open-ended consulting.

1. Solution Overview — Microsoft 365 ↔ Google Workspace Coexistence

During any multi-month migration between **Microsoft 365** and **Google Workspace**, the two ecosystems run side by side — and users expect mail, calendars, and the directory to *just work* across both. This solution designs and implements **coexistence**: the set of mail-routing, calendar free/busy, and directory-sync controls that make Microsoft and Google behave as one organization until the migration completes and the legacy side is retired.

It is authored for a **competent administrator** who wants a repeatable, click-to-implement path. Every connector, endpoint, role account, and organizational unit is **named once and reused verbatim** across the requirements, implementation, testing, troubleshooting, and the five supporting scripts, so the design lines up end to end.

The deployment model is the standard **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** with explicit pass/fail gates and a documented rollback for every production change — mail flow, DNS, and identity changes are high-impact and are always piloted before fleet-wide rollout.

The three pillars of coexistence

Coexistence is not one setting; it is three independent interop channels that together make the split organization feel unified. Each pillar is configured, piloted, and validated on its own.

Pillar 1 — Mail routing / split delivery

While a domain (e.g. `contoso.com`) is shared between both platforms, inbound mail lands on one **primary** system (whichever holds the domain's MX record) and is **relayed** to the other for recipients that don't exist locally. This is *split delivery*.

- **MX-primary platform** — the platform the domain's **MX record** points to receives all inbound

mail first. Common patterns during a Google-to-M365 migration keep Google as MX-primary early, then flip to M365 as mailboxes move.

- **Relay for unknown recipients** — on each platform the shared domain is set to a **non-authoritative

/ internal-relay** mode so that a message for a recipient that isn't hosted locally is forwarded to the other platform rather than bounced (NDR).

- On **Exchange Online**: the shared domain's **Accepted Domain** type is set to **Internal relay**,

and an **outbound connector** (named `Coexistence – EXO→Google Connector`) relays unknown recipients to Google's smart host (`aspmx.l.google.com`).

- On **Google Workspace**: **Routing** → **Recipient address map / Split delivery** (Admin console →

Apps → **Google Workspace** → **Gmail** → **Routing**) is configured so mail for recipients not hosted in Google is routed to the M365 host (the domain's `*.mail.protection.outlook.com` endpoint).

- **Authentication** — **SPF** must authorize *both* sending platforms; **DKIM** is signed

separately on each; **DMARC** is kept in a monitoring-friendly policy during the transition so a misconfigured relay does not cause silent drops.

The result: a sender anywhere reaches the right recipient regardless of which platform hosts that mailbox on any given day.

Pillar 2 — Free/busy calendar interop

Users on both platforms must see each other's **availability** when scheduling. This is delivered by **Google Calendar Interop** (Admin console → **Apps** → **Google Workspace** → **Calendar** → **Calendar Interop management**) paired with Exchange-side availability sharing:

- **Google → Exchange direction** — Google Calendar queries Exchange availability through an

Exchange endpoint (Microsoft 365 Graph API endpoint, or EWS for legacy) using one or more **Exchange role accounts** (named `svc-interop-gcal@contoso.com`). Google users then see Exchange users' free/busy.

- **Exchange → Google direction** — Exchange queries Google availability through a dedicated

Google role account (named `svc-interop-exo@contoso.com`) referenced in Calendar Interop. Exchange/Outlook users then see Google users' free/busy.

- **Detail level** — lookups return **availability only** by default; **event details** (title/location)

can be enabled per direction, and event-level privacy is respected on both sides.

The result: Outlook and Google Calendar both resolve availability for anyone in the org, so meeting scheduling doesn't break during the split.

Pillar 3 — Directory sync

A single, consistent **Global Address List / directory** across both platforms lets users find and address each other. The identity **source of truth** (on-prem AD, Entra ID, or an HR/IGA system) projects users and groups into *both* ecosystems:

- **Into Google Workspace** — **Google Cloud Directory Sync (GCDS)** performs a **one-way** sync

from LDAP/Active Directory into Google (users, groups, and shared contacts). GCDS never writes back to the source. Mail-enabled contacts for M365-only recipients are provisioned as **shared contacts** so they appear in Google's directory.

- **Into Microsoft 365 / Entra** — users are provisioned from the same source (Entra Connect / cloud

sync from AD, or native Entra where Entra is the source). Google-only recipients are represented in Exchange as **mail contacts / mail users** so they appear in the GAL and route correctly.

- **Convention** — objects synced for coexistence carry a recognizable naming/attribute marker (e.g.

a custom attribute or OU `Coexistence – Synced`) so they can be reported on and cleanly removed at consolidation.

The result: one directory experience — everyone can look up and address everyone, on either platform.

How the pillars work together

Pillar	Google side object	Microsoft side object	Makes possible
Mail routing	Split-delivery routing rule	Internal-relay domain + `Coexistence – EXO→Google Connector`	Mail reaches the right mailbox on either platform
Free/busy	Calendar Interop + role account `svc-interop-exo@contoso.com`	Exchange endpoint + role account `svc-interop-gcal@contoso.com`	Cross-platform availability in scheduling
Directory	GCDS + shared contacts	Entra provisioning + mail contacts	One unified directory / GAL

Together they produce a **unified experience**: mail routes, calendars resolve, and the directory stays in sync — for as long as the two platforms coexist.

Scope, boundaries, and lifecycle

• **In scope:** design and configuration of mail routing/split delivery, free/busy interop, and directory sync; a pilot-first rollout with gates; validation tooling; and a documented rollback.

• **Out of scope (deferred to vendor docs):** bulk **data migration** (mailbox/Drive/Chat content movement) performed by third-party migration tools — coexistence keeps the org working *while* that migration runs, but is a separate workstream. Third-party enterprise apps are integrated at the pattern level only.

• **Vendor boundary:** the **Google Admin console** and the **Microsoft / Entra** sides are authored at click-by-click detail from the vendors' current public documentation. Where a capability is preview/rolling-out (e.g. specific Interop endpoint options), it is flagged **"verify in your tenant."**

• **Lifecycle:** coexistence is a **transition state**. It is stood up pilot-first, expanded to the fleet, operated for the migration window, and then **decommissioned** in a controlled way once consolidation retires the legacy platform (see `07-best-practices.md`).

> ***Cited sources.*** *Microsoft Learn — *Manage accepted domains* / *Set up connectors to route mail* (internal-relay + smart-host routing to `aspmx.l.google.com`); Google Workspace Admin Help — *About Calendar Interop*, *Allow Calendar users to see Exchange availability* (role accounts, endpoints, availability vs. event details); Google Workspace Admin Help — *About Google Cloud Directory Sync* (one-way LDAP→Google sync). Preview/rolling-out endpoint options are flagged "verify in your tenant."*

2. Requirements — Microsoft 365 ↔ Google Workspace Coexistence

Verify **every** requirement below before starting the implementation runbook (`03-implementation.md`). Each requirement is stated as **Description** → **Why required** → **How to verify** → **How to implement if missing**. These are in addition to the Google-category **global requirements**

(`global-requirements-google.md`), which cover the base tenants, DNS control, identity source of truth, pilot scope, and governance stakeholders.

Naming used throughout (reuse verbatim):

- Shared vanity domain: `contoso.com`
- Google role account (used by Exchange to read Google availability): `svc-interop-exo@contoso.com`
- Exchange role account (used by Google to read Exchange availability): `svc-interop-gcal@contoso.com`
- Exchange outbound relay connector: `Coexistence – EXO→Google Connector`
- Directory-sync marker OU/group: `Coexistence – Synced`
- Pilot group: `Coexistence – Pilot Members`

R1 — Administrative access to both platforms

- **Description:** A **Google Workspace super-admin** account **and** a Microsoft 365 account with the roles the steps name (Exchange admin for mail flow/connectors; Global Admin or the least-privilege equivalent for domain, Calendar-endpoint, and directory work).
- **Why required:** Every pillar changes configuration on both sides; you cannot configure split delivery, Interop, or directory sync from one platform alone.
- **How to verify:** Sign in to **admin.google.com** as super-admin and to **admin.microsoft.com** / **admin.exchange.microsoft.com** with the required role.
- **How to implement if missing:** Obtain the roles from each tenant owner before starting; record them in change-control.

R2 — Shared/verified domain on both platforms

- **Description:** The domain being coexisted (`contoso.com`) is **added and verified** on *both* Google Workspace and Microsoft 365.
- **Why required:** Split delivery requires the same SMTP domain to be a recognized (accepted) domain on each platform so each can decide "mine, or relay to the other."
- **How to verify:** Google Admin console → **Account** → **Domains** → **Manage domains** shows the domain verified; Microsoft 365 admin center → **Settings** → **Domains** shows it verified; Exchange admin center → **Mail flow** → **Accepted domains** lists it.
- **How to implement if missing:** Add and verify the domain (TXT/CNAME) on whichever platform is missing it. Do **not** yet change its authoritative/relay type — that is a Configure-phase step.

R3 — DNS control for MX, SPF, DKIM, DMARC

• **Description:** Ability to edit the authoritative DNS zone for `contoso.com` — specifically **MX** (routing primary), **SPF** (authorize both senders), **DKIM** (one selector per platform), and **DMARC**.

• **Why required:** MX decides which platform is inbound-primary; SPF/DKIM/DMARC must accommodate two sending platforms or legitimate mail will fail authentication and be dropped.

• **How to verify:** Confirm write access to the zone; record the **current** MX/SPF/DKIM/DMARC as the rollback baseline (`GWCOEX-Discovery.ps1` captures the M365 side; capture the Google/DNS side too).

• **How to implement if missing:** Coordinate with the DNS owner; stage all changes under change control with lowered TTLs first (see `07-best-practices.md`).

R4 — Decided MX-primary platform and routing direction

• **Description:** A documented decision on which platform is **MX-primary** during coexistence and therefore which platform **relays** unknown recipients to the other (and when, if ever, MX flips).

• **Why required:** Split delivery is asymmetric — exactly one platform receives inbound first and relays the remainder. The relay connector and Google routing rule are built for that direction.

• **How to verify:** Confirm the current MX target and the agreed coexistence direction are written down and approved.

• **How to implement if missing:** Decide with the migration owners (typical early state: Google MX-primary relaying unknowns to M365; late state: the reverse). Record it; the runbook builds to it.

R5 — Exchange side ready for internal-relay mail flow

• **Description:** On Exchange Online, the ability to set the shared domain's **Accepted Domain** type to **Internal relay** and to create an **outbound connector** to Google's smart host.

• **Why required:** Without internal-relay + a connector, Exchange NDRs mail addressed to Google-hosted recipients instead of relaying it.

• **How to verify:** Exchange admin center → **Mail flow** → **Accepted domains** (can you edit the type?) and **Mail flow** → **Connectors** (can you create a connector?). `GWCOEX-Discovery.ps1` reports the current accepted-domain types and connectors.

• **How to implement if missing:** Grant the Exchange admin role; confirm no conflicting existing connector already claims the domain.

R6 — Google side ready for split-delivery routing

• **Description:** Access to **Apps** → **Google Workspace** → **Gmail** → **Routing** (and **Hosts**) in the Google Admin console to define a **split-delivery** rule that routes unknown recipients to the M365 host.

• **Why required:** This is the Google-side mirror of internal relay — it forwards mail for recipients not hosted in Google to Exchange Online rather than bouncing.

• **How to verify:** Confirm the Gmail **Routing** and **Hosts** pages are accessible; note the M365 inbound host (`contoso-com.mail.protection.outlook.com`).

• **How to implement if missing:** Assign a role that includes Gmail settings; obtain the domain's M365 `*.mail.protection.outlook.com` value from Microsoft 365 admin center → **Settings** → **Domains**.

R7 — Free/busy interop prerequisites (both directions)

• **Description:** The building blocks for **Calendar Interop**:

• a dedicated **Google role account** `svc-interop-exo@contoso.com` (standard user, Calendar enabled, no personal calendar use) that Exchange uses to read Google availability;

• a dedicated **Exchange role account** `svc-interop-gcal@contoso.com` (standard mailbox, password non-expiring if basic auth) that Google uses to read Exchange availability;

• an **Exchange endpoint** reachable by Google — **Microsoft 365 Graph API** endpoint (app registration with the required Calendars permission) or the legacy **EWS URL** (`https://outlook.office365.com/EWS/Exchange.asmx`) over **port 443** with a public-CA TLS certificate.

• **Why required:** Interop is bidirectional and each direction authenticates with its own role account; Google must reach the Exchange endpoint to resolve availability.

• **How to verify:** Confirm both role accounts exist and are licensed/enabled; from the internet, confirm the Exchange endpoint responds on 443. **Graph-API endpoint availability can be preview/rolling-out — verify the endpoint types offered in your tenant.**

• **How to implement if missing:** Create the two role accounts; register the app (Graph) or confirm EWS reachability; apply availability sharing (`Set-MailboxCalendarFolder … -SharingPermissionFlags` / detail level) on the Exchange side.

R8 — Directory-sync mechanism and host

• **Description:** A **directory-sync** path from the identity source of truth into **both** platforms:

Google Cloud Directory Sync (GCDS) installed on a supported host with LDAP access to AD (for the Google side), and Entra provisioning (Entra Connect / cloud sync, or native Entra) for the Microsoft side. A dedicated **Google provisioning admin** (e.g. `svc-gcds@contoso.com`) for GCDS OAuth.

• **Why required:** A unified directory/GAL across platforms depends on the same source projecting users, groups, and cross-platform contacts into each ecosystem. GCDS is **one-way** (never writes to AD).

• **How to verify:** Confirm the source of truth (per global requirement 4), that GCDS's host can reach LDAP (port 389/636), and that the Google provisioning admin can authorize GCDS. `GWCOEX-Discovery.ps1` reports the M365/Entra recipient and contact counts.

• **How to implement if missing:** Stand up a GCDS host (separate from the domain controller is recommended), create the provisioning admin, and define which OUs/groups sync. Decide how each platform represents the *other* platform's recipients (Google **shared contacts** ↔ Exchange **mail contacts / mail users**).

R9 — A pilot scope and change control

• **Description:** A named **pilot group** (`Coexistence – Pilot Members`) spanning both platforms (some Google-hosted, some M365-hosted users) and a change-control window for mail-flow/DNS/identity changes.

• **Why required:** The model is **Assess** → **Configure** → **Pilot** → **Validate** → **Expand**; mail-flow and DNS changes are high-impact and must be proven on a small, representative set before fleet rollout.

• **How to verify:** Confirm the pilot group exists with members on both platforms and that an approval path/window is available.

• **How to implement if missing:** Create `Coexistence – Pilot Members`; include at least a few users on each platform plus one resource/room and one shared mailbox equivalent; agree the change window with the owners.

All prerequisites met? Proceed to `03-implementation.md`. Any gap — especially R3 (DNS), R4 (routing direction), or R9 (pilot) — is a stop condition: coexistence changes production mail flow and must not begin until the baseline is captured and the pilot and rollback are in place.

Global Requirements — Google Workspace & Gemini Solutions

These requirements apply to **every** solution in the **Google** category. They are the common foundation each Google Workspace / Gemini solution assumes is already in place (or stands up as part of its own Requirements section). Verify each before starting a solution's implementation steps.

> ***Vendor boundary (applies to every Google solution).*** *The **Google Admin console** (admin.google.com), **Google Cloud / Cloud Identity**, and the **Microsoft / Entra** sides are authored at exact, current click-by-click detail from the vendors' public documentation. **Gemini Enterprise** capabilities that are in preview or rolling out are **cited or flagged "verify in your tenant"** — never invented. **Third-party migration tools** (e.g. for mail/Drive/Chat data movement) and **third-party enterprise apps** (ServiceNow, SAP, Databricks, etc.) are integrated at the pattern/architectural level;*

their internal console steps are deferred to the vendor's own docs.

1. Google Workspace / Cloud Identity tenant

• **Description:** A provisioned Google Workspace (or Cloud Identity) tenant with a **super-admin** account for configuration, and the appropriate Workspace edition/licensing for the capability in scope (e.g. Enterprise Plus for Vault/CAA/DLP, Gemini/AI add-on where required).

• **Why required:** All configuration, policy, and API work is performed against this tenant.

• **How to verify:** Sign in to **admin.google.com** as a super-admin; confirm edition and license counts under **Billing / Subscriptions**.

• **How to implement if missing:** Provision the tenant and licenses through Google or an authorized reseller before starting.

2. Source Microsoft 365 / Entra environment (for migration & coexistence)

• **Description:** For migration/coexistence/identity solutions, administrative access to the source **Microsoft 365** tenant and **Microsoft Entra ID** (Global Administrator or the least-privilege roles the solution names).

• **Why required:** Migration, directory sync, SSO federation, and coexistence read from and change the Microsoft side.

• **How to verify:** Confirm access to the M365 admin center and the Entra admin center with the required role.

• **How to implement if missing:** Obtain the role from the M365/Entra tenant owner.

3. DNS control for the routed domains

• **Description:** Ability to create/modify **DNS records** (MX, SPF, DKIM, DMARC, TXT, CNAME) for the domains being migrated, coexisting, or verified.

• **Why required:** Domain verification, mail routing/coexistence, and authentication all depend on DNS changes.

• **How to verify:** Confirm access to the authoritative DNS zone for each domain.

• **How to implement if missing:** Coordinate with whoever controls DNS; stage changes under change control (mail-flow changes are high-impact).

4. Identity & provisioning source of truth

• **Description:** A defined **identity source of truth** (on-prem AD, Entra ID, or an HR/IGA system) and the sync mechanism (e.g. **Google Cloud Directory Sync (GCDS)** or SCIM-based provisioning).

• **Why required:** Consistent users/groups across ecosystems underpin SSO, licensing, and access.

- **How to verify:** Confirm which directory is authoritative and how objects flow into Google.
- **How to implement if missing:** Define the source of truth and provisioning model before configuring SSO or migrating.

5. A pilot scope (OU / group) and change control

- **Description:** A **pilot organizational unit or group** to validate on before fleet-wide rollout, and a **change-control** process for production changes (especially mail flow, DNS, identity, and DLP).
- **Why required:** Every solution follows an **Assess** → **Configure** → **Pilot** → **Validate** → **Expand** model; the pilot proves the design and change control protects production.
- **How to verify:** Confirm a named pilot OU/group exists and a change window/approval path is available.
- **How to implement if missing:** Create a pilot OU/group and agree the change-control process with the owners before making production changes.

6. Security, compliance & governance stakeholders

- **Description:** Engagement with the **security (CISO), legal/eDiscovery, and data-governance** stakeholders whose policies the configuration must satisfy (DLP, retention, Vault/Purview, and any regulatory regime such as ITAR/DFARS/Assured Controls where applicable).
- **Why required:** Data governance, eDiscovery, DLP, and regulated-workload settings must align to enterprise policy and legal requirements, not just technical defaults.
- **How to verify:** Confirm the applicable policies and named stakeholders/approvers.
- **How to implement if missing:** Identify the owners and obtain the policy/compliance requirements before configuring governance controls.

*The deployment model across every Google solution is the same: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand**, with explicit pass/fail gates and a documented rollback for production changes.*

3. Implementation — Microsoft 365 ↔ Google Workspace Coexistence

This is the ordered runbook. Execute the phases in order: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand**. Do **not** advance a phase until its exit gate (see `05-testing.md`) passes. Every object is **named once and reused verbatim** in later steps, testing, troubleshooting, and the scripts.

Conventions used below

- Shared vanity domain: `contoso.com` · M365 inbound host: `contoso-com.mail.protection.outlook.com`

- Google smart host: `aspmx.l.google.com`
- Google role account (Exchange → reads Google availability): `svc-interop-exo@contoso.com`
- Exchange role account (Google → reads Exchange availability): `svc-interop-gcal@contoso.com`
- GCDS provisioning admin: `svc-gcds@contoso.com`
- Exchange outbound relay connector: `Coexistence – EXO→Google Connector`
- Google split-delivery host: `Coexistence – M365 Host` · routing rule: `Coexistence – Split Delivery`
- Directory-sync marker OU/group: `Coexistence – Synced` · Pilot group: `Coexistence – Pilot Members`

> *Replace `contoso.com` and the host values with your real values, but keep the naming **pattern** identical so downstream steps and scripts line up.*

Phase 1 — Assess

1.1 Confirm access to both platforms

1. Sign in to **admin.google.com** (super-admin) and **admin.exchange.microsoft.com** / **admin.microsoft.com** (Exchange admin + Global Admin or least-privilege equivalent) — **R1**.

1.2 Capture the coexistence-readiness baseline

1. On an operator workstation, install the Exchange Online module: `Install-Module ExchangeOnlineManagement -Scope CurrentUser`.
1. Run `scripts\GWCOEX-Discovery.ps1` against the M365 tenant. It exports CSVs of: accepted domains and their **type** (authoritative vs internal relay), inbound/outbound **connectors**, the domain's **MX / SPF / DKIM / DMARC** as seen in DNS, **mail contacts / mail users** (Google-side representation), and any existing **Calendar Interop / organization sharing** config.
1. Separately record the current **Google** side (Domains, Gmail Routing/Hosts, Calendar Interop management state) and the **DNS** MX/SPF/DKIM/DMARC. Save all of this as the **rollback baseline**.

Exit gate: complete readiness baseline captured on both platforms and DNS; MX-primary direction (**R4**) confirmed and written down. (Test **T1**.)

Phase 2 — Configure

Configure the three pillars. Build each for the **pilot scope first**; expansion is Phase 5.

2.1 Mail routing / split delivery

2.1a Exchange side — internal relay + connector to Google

1. In **Exchange admin center** → **Mail flow** → **Accepted domains**, open `contoso.com` and set the type

to **Internal relay**. (This tells Exchange to *relay* unknown recipients instead of NDR.)

1. In **Mail flow** → **Connectors** → **Add a connector**, create an **outbound** connector

`Coexistence – EXO→Google Connector`: From **Office 365** → To **Partner organization**; route by **domain** `contoso.com`; delivery via **smart host** `aspmx.l.google.com` (or the domain's full Google MX target); require **TLS** with a trusted CA subject. Save.

• You may scope relay to the pilot initially by using a routing subdomain or a transport rule that

limits the connector to `Coexistence – Pilot Members` during Phase 3.

1. Confirm mail for a Google-hosted pilot recipient now relays out of Exchange rather than NDRing

(validated in Phase 4).

2.1b Google side — split-delivery routing to M365

1. In **Google Admin console** → **Apps** → **Google Workspace** → **Gmail** → **Hosts**, add a host

`Coexistence – M365 Host` = `contoso-com.mail.protection.outlook.com`, port **25**, require TLS.

1. In **Apps** → **Google Workspace** → **Gmail** → **Routing**, add a **Recipient address map / split-delivery**

rule `Coexistence – Split Delivery`: for envelope recipients **not** hosted in Google, route to `Coexistence – M365 Host`. Keep the rule scoped to the pilot OU during Phase 3.

1. This is the mirror of internal relay: Google delivers to its own mailboxes and forwards the rest

to Exchange Online.

2.1c Authentication (DNS)

1. **SPF:** publish a single TXT that authorizes **both** senders, e.g.

`v=spf1 include:_spf.google.com include:spf.protection.outlook.com -all`.

1. **DKIM:** enable and publish a **separate selector** for each platform (Google: `google` selector;

M365: the two `selector1/2._domainkey` CNAMEs). Both sign independently.

1. **DMARC:** during the transition keep a monitoring-friendly policy (`p=none` with `rua=` reporting,

or `quarantine` with a wide `pct`) so a relay misstep is *visible*, not silently dropped. Tighten only after Validate.

2.2 Free/busy calendar interop

2.2a Create the two role accounts

1. In **Google Admin console** → **Directory** → **Users**, create `svc-interop-exo@contoso.com` — a

standard, Calendar-enabled user Exchange will use to read Google availability. Do not use it as a person's mailbox.

1. In **Microsoft 365 admin center** → **Users**, create `svc-interop-gcal@contoso.com` — a standard

licensed mailbox Google will use to read Exchange availability. If the endpoint uses basic auth, set the password to **non-expiring** and exclude the account from blocking Conditional Access.

2.2b Google → Exchange (Google users see Exchange availability)

1. In **Google Admin console** → **Apps** → **Google Workspace** → **Calendar** → **Calendar Interop management**,

check **"Allow Google Calendar to display Exchange users' availability."**

1. Click **Add an Exchange endpoint** and choose the endpoint type your tenant offers —

Microsoft 365 (Graph API) (provide **Tenant ID**, **Application (client) ID**, **client secret** from an app registration granted the required Calendars permission) **or** legacy **EWS** (`https://outlook.office365.com/EWS/Exchange.asmx` with role account `svc-interop-gcal@contoso.com`). *(Graph endpoint availability may be preview/rolling-out — verify in your tenant.)*

1. Optionally enable **Show event details** (title/location) instead of availability-only; event-level

privacy is respected.

2.2c Exchange → Google (Exchange users see Google availability)

1. Still in **Calendar Interop management**, set the **Google role account**

`svc-interop-exo@contoso.com` that Exchange will authenticate as when reading Google availability.

1. On the Exchange side, configure organization availability sharing so Outlook resolves the Google

users. Where event-detail visibility is wanted, raise the calendar detail level, e.g. `Set-MailboxCalendarFolder <user>:\Calendar -DetailLevel LimitedDetails` for pilot users (default is availability-only).

1. Ensure the Exchange endpoint is reachable from Google on **port 443** with a public-CA TLS cert

(allowlist Google's Interop egress ranges if you block inbound).

2.3 Directory sync (unified GAL)

2.3a Google side — GCDS from AD/LDAP

1. In **Google Admin console** → **Directory** → **Users**, create the provisioning admin

`svc-gcds@contoso.com` with directory + shared-contacts privileges.

1. Install **GCDS** on a supported host (separate from the domain controller is recommended) with

LDAP access to AD (389/636).

1. In **GCDS Configuration Manager**:

• **Google Domain Configuration** → domain `contoso.com`; **Authorize** with `svc-gcds@contoso.com`.

• **LDAP Configuration** → server type **MS Active Directory**, base DN, and a read account; **Test

connection**.

• **User Accounts / Groups / Shared Contacts** → search rules that select the **`Coexistence – Synced` **OU/group. Provision** shared contacts** for M365-only recipients so they appear in Google's directory.

• **Sync** → **Clear cache** → **Simulate sync**; review **Proposed changes**; confirm no unexpected deletions, then run once for the pilot.

1. GCDS is **one-way** — it never modifies AD.

2.3b Microsoft side — provision from the same source

1. Ensure users are provisioned into **Entra/Exchange** from the same source of truth (Entra Connect / cloud sync from AD, or native Entra).

1. Represent **Google-only** recipients in Exchange as **mail contacts / mail users** (so they appear in the GAL and route via 2.1a). Tag coexistence-synced objects with the `Coexistence – Synced` marker for later cleanup.

Exit gate: all three pillars configured and scoped to the pilot; DNS auth (SPF/DKIM/DMARC) in place. (Test **T2**.)

Phase 3 — Pilot

1. Populate `Coexistence – Pilot Members` with representative users on **both** platforms plus one resource/room and one shared mailbox equivalent.

1. Confirm the mail relay, split-delivery rule, and Interop endpoint are scoped to the pilot (subdomain, transport rule, or OU) so production users are unaffected.

1. Run `scripts\GWCOEX-Set-ScopeTargeting.ps1` to (re)point the coexistence scope at `Coexistence – Pilot Members`. It defaults to `-WhatIf` — review the planned scope change, then re-run with `-Apply`.

Exit gate: coexistence live for the pilot only; production unchanged. (Test **T3**.)

Phase 4 — Validate

Run `scripts\GWCOEX-Validation.ps1` for each pilot user; it checks the three pillars per user:

1. **Mail routing** — send a probe from Google→M365 and M365→Google for pilot recipients; confirm delivery (no NDR) and correct headers (relayed via the connector / split-delivery host).

1. **Free/busy** — confirm a Google user resolves an Exchange user's availability *and* an Outlook user resolves a Google user's availability (both directions).

1. **Directory** — confirm each pilot user appears in the **other** platform's directory/GAL (Google shared contact ↔ Exchange mail contact) and resolves when addressed.

Run `scripts\GWCOEX-Report.ps1` to produce a **coexistence coverage** report (per user: mail-route OK, free/busy both-ways OK, directory-present OK) with an overall coverage percentage.

If a check fails, use `06-troubleshooting.md`; if remediation is needed, run `scripts\GWCOEX-Remediation.ps1` (defaults to `-WhatIf` — review, then `-Apply`).

Exit gate: 100% of pilot users pass all three pillars, both directions. (Test **T4**.)

Phase 5 — Expand

1. Widen each pillar's scope from `Coexistence – Pilot Members` to the full fleet:

• remove the pilot scoping on the **connector / transport rule** and the Google **split-delivery rule** (or point them at the whole domain/OU);

• broaden the **GCDS** search rules to all coexistence OUs and schedule GCDS on its interval;

• confirm the Interop endpoint covers all users.

1. Re-run `GWCOEX-Set-ScopeTargeting.ps1 -Scope AllUsers` (defaults to `-WhatIf`; re-run with `-Apply`).

1. Re-run `GWCOEX-Validation.ps1` on a broad sample and `GWCOEX-Report.ps1` for fleet coverage.

2. Only after coverage is proven, tighten **DMARC** to an enforcing policy.

Exit gate: fleet coverage report green; DMARC enforced; rollback plan current. (Test **T5**.)

Rollback (any phase)

Because every change is captured in the Phase 1 baseline, rollback is deterministic:

• Revert the **Accepted Domain** type to its prior value and **disable** `Coexistence – EXO→Google Connector`.

• Disable the Google `Coexistence – Split Delivery` routing rule.

• Uncheck **Calendar Interop** endpoints.

• **Pause GCDS** (do not delete synced objects until consolidation).

• Restore prior **MX/SPF/DKIM/DMARC** DNS if they were changed.

Coexistence is a transition state — see `07-best-practices.md` for the controlled **decommission** at consolidation.

5. Testing & Validation — Microsoft 365 ↔ Google Workspace Coexistence

Each phase in `03-implementation.md` has a **pass/fail exit gate**. Do not advance until the gate's test passes. Tests use the same named objects as the runbook (`Coexistence – Pilot Members`, `Coexistence – EXO→Google Connector`, `svc-interop-*`, etc.). The scripts automate the per-user checks; this file defines what "pass" means.

Test accounts: at least one **Google-hosted** and one **M365-hosted** pilot user, plus one resource/room and one shared mailbox equivalent, all in `Coexistence – Pilot Members`.

T1 — Assess gate (readiness baseline)

Goal: the coexistence-readiness baseline is complete on both platforms and DNS.

Check	How	Pass
M365 baseline	`GWCOEX-Discovery.ps1` ran; CSVs present	Accepted-domain types, connectors, contacts, and Interop state exported
DNS baseline	Inspect MX/SPF/DKIM/DMARC for `contoso.com`	Current records recorded as rollback baseline
Google baseline	Domains, Gmail Routing/Hosts, Calendar Interop management state noted	State recorded
Direction decided	Review R4 decision	MX-primary platform + relay direction written down and approved

Fail → capture the missing baseline before any Configure step.

T2 — Configure gate (all three pillars built for pilot)

Pillar	Check	Pass
Mail	EXO accepted domain = **Internal relay**; `Coexistence – EXO→Google Connector` exists; Google `Coexistence – Split Delivery` rule + `Coexistence – M365 Host` exist	Both relay paths configured (pilot-scoped)
Free/busy	Exchange endpoint added in Calendar Interop; role accounts `svc-interop-exo@contoso.com` and `svc-interop-gcal@contoso.com` set	Both directions configured

Pillar	Check	Pass
Directory	GCDS simulate-sync shows expected users/groups/shared contacts; EXO mail contacts for Google-only recipients exist	Directory projection ready both ways
Auth	SPF authorizes both senders; DKIM selectors per platform; DMARC monitoring policy	DNS auth in place

Fail → fix the specific pillar in `03-implementation.md` §2.x before piloting.

T3 — Pilot gate (scoped to pilot only)

Check	How	Pass
Scope correct	`GWCOEX-Set-ScopeTargeting.ps1` (default `-WhatIf`) plan shows scope = `Coexistence – Pilot Members`	Only pilot users in scope
Production untouched	Send a normal message to a non-pilot production user	Delivered exactly as before coexistence

Fail → re-scope before validating; never validate with production in scope.

T4 — Validate gate (pilot passes all pillars, both directions)

Run `GWCOEX-Validation.ps1` per pilot user, then `GWCOEX-Report.ps1`.

Pillar	Test	Pass
Mail G→M	Google-hosted user emails an M365-hosted pilot user	Delivered, no NDR; header shows relay/split-delivery path
Mail M→G	M365-hosted user emails a Google-hosted pilot user	Delivered, no NDR; relayed via `Coexistence – EXO→Google Connector`
Free/busy G→X	Google user views an Exchange user's availability	Busy blocks resolve
Free/busy X→G	Outlook user views a Google user's availability	Busy blocks resolve
Directory G	Google-only recipient appears in EXO GAL (mail contact)	Resolves when addressed
Directory M	M365-only recipient appears in Google directory (shared contact)	Resolves when addressed

Pass gate: 100% of pilot users green on all six rows. **Fail** → `06-troubleshooting.md`, then `GWCOEX-Remediation.ps1` (default `-WhatIf`).

T5 — Expand gate (fleet coverage)

Check	How	Pass
Scope widened	`GWCOEX-Set-ScopeTargeting.ps1 -Scope AllUsers` plan	Whole fleet in scope
Fleet coverage	`GWCOEX-Report.ps1` across a broad sample	Coverage ≥ agreed threshold (target 100%)
GCDS scheduled	GCDS running on interval	Automated sync healthy, no unexpected deletes
DMARC enforced	DNS shows enforcing DMARC	Enforcement on only after coverage proven

Fail → do not tighten DMARC or declare done; remediate and re-report.

Regression / ongoing checks

- **Weekly:** `GWCOEX-Report.ps1` for coverage drift; investigate any user that drops a pillar.
- **After any DNS or connector change:** re-run mail-routing rows (T4 Mail G→M / M→G).
- **After GCDS schema/OU changes:** re-run directory rows (T4 Directory G / M).
- **Before consolidation cutover:** full T4 sweep, then follow the decommission steps in `07-best-practices.md`.

6. Troubleshooting — Microsoft 365 ↔ Google Workspace Coexistence

Symptom → likely cause → fix, organized by pillar. Reuse the named objects from `03-implementation.md`. When a fix requires a change, prefer `GWCOEX-Remediation.ps1` (default `-WhatIf`) so you can review before applying. Always re-run the matching `05-testing.md` row after a fix.

Mail routing / split delivery

*Mail to a valid recipient on the *other* platform bounces (NDR: recipient not found)*

- **Cause:** the shared domain is still **Authoritative** on the receiving platform, so it rejects recipients it doesn't host instead of relaying.
- **Fix:** on Exchange, set `contoso.com` Accepted Domain to **Internal relay**; on Google, ensure the

`Coexistence – Split Delivery` routing rule is enabled and targets `Coexistence – M365 Host`. Verify the outbound path (`Coexistence – EXO→Google Connector` / the Google host) exists and is on.

Relayed mail loops or bounces with "too many hops"

- **Cause:** *both* platforms are relaying the same domain to each other with no authoritative owner for

a nonexistent address — the message ping-pongs.

- **Fix:** confirm exactly **one MX-primary** (R4). A message for a truly nonexistent recipient must

bounce cleanly on the MX-primary; it should not be relayed onward. Re-check the split-delivery scope so only *unknown-locally* recipients relay.

Relayed mail is marked spam or fails authentication at the far platform

- **Cause: SPF** doesn't authorize both senders, or **DKIM** isn't signing on the relaying platform.
- **Fix:** publish one SPF TXT with `include:_spf.google.com include:spf.protection.outlook.com`; enable

a **separate DKIM selector** per platform. Audit SPF for the **10-lookup limit**. Keep **DMARC** at a monitoring policy until both align.

Some pilot users route, others NDR

- **Cause:** the relay/split-delivery rule is scoped to the pilot but the failing user isn't in

`Coexistence – Pilot Members` (or the connector transport rule excludes them).

- **Fix:** confirm scope with `GWCOEX-Set-ScopeTargeting.ps1` (default `-WhatIf`); add the user or widen

the scope, then re-test.

Relayed mail is delivered in cleartext / TLS warnings

- **Cause:** the connector or Google host isn't requiring TLS.
- **Fix:** enable **Require TLS** on `Coexistence – EXO→Google Connector` and the Google host

`Coexistence – M365 Host`.

Free/busy calendar interop

Google user sees an Exchange user as blank / "no information"

- **Cause:** the **Exchange endpoint** or `svc-interop-gcal@contoso.com` role account is

misconfigured, or the endpoint is unreachable.

- **Fix:** in **Calendar Interop management**, re-check the endpoint (Graph app credentials or the EWS

URL `https://outlook.office365.com/EWS/Exchange.asmx`) and role account. Confirm the endpoint answers on **443** with a valid public-CA cert; if inbound is filtered, allowlist Google's Interop egress ranges.

Outlook user sees a Google user as blank

• **Cause:** the **Google role account** `svc-interop-exo@contoso.com` isn't set in Interop, is Calendar-disabled, or the Exchange-side org availability sharing isn't pointed at Google.

• **Fix:** confirm the Google role account is a standard, Calendar-enabled user set in Interop; verify the Exchange organization availability/sharing configuration references it.

Availability resolves but no event titles/locations

• **Cause:** detail level is **availability-only** (the default).

• **Fix:** enable **Show event details** in Interop (Google→Exchange) and raise the Exchange calendar detail level (e.g. `Set-MailboxCalendarFolder <user>:\Calendar -DetailLevel LimitedDetails`) where policy allows. Event-level privacy is still respected.

Interop works for some users, not others

• **Cause:** the target user has a conflicting personal Google account on the domain, an Exchange mailbox issue, or isn't in scope yet.

• **Fix:** ensure interop-target users have no conflicting personal Google account and a healthy mailbox; confirm they're in the current scope.

Endpoint type "Microsoft 365 (Graph API)" not available

• **Cause:** the Graph-API endpoint option is **preview/rolling-out** in your tenant.

• **Fix: verify in your tenant**; if unavailable, use the legacy **EWS** endpoint with modern-auth where supported, and revisit when Graph is offered.

Directory sync

A user on the other platform doesn't appear in the address book/GAL

• **Cause:** the cross-platform representation is missing — no Google **shared contact** for an M365-only user, or no Exchange **mail contact / mail user** for a Google-only user.

• **Fix:** confirm GCDS provisions shared contacts for M365-only recipients and that EXO mail contacts exist for Google-only recipients (create via `GWCOEX-Remediation.ps1`, default `-WhatIf`). Confirm the user is in the `Coexistence – Synced` OU/group.

GCDS simulate-sync proposes mass deletions

• **Cause:** an over-broad or wrong **search rule / base DN**, so GCDS thinks synced objects should be removed.

• **Fix: do not run the real sync.** Correct the LDAP search rules / base DN to the intended `Coexistence – Synced` scope; re-**Simulate** until **Proposed changes** are as expected.

Directory in Google is stale / out of date

• **Cause:** GCDS isn't scheduled, or its last run failed.

• **Fix:** confirm GCDS is scheduled on its interval and the last run succeeded; check the GCDS logs and LDAP connectivity (389/636).

Duplicate or conflicting entries appear

• **Cause:** two systems are authoring the same user (the GAL diverges).

• **Fix:** enforce a **single source of truth**; ensure only one provisioning path writes each user, and that shared contacts/mail contacts are only used for the *other* platform's recipients.

Cross-cutting

`GWCOEX-*.ps1` *fails to connect*

• **Cause:** the Graph SDK / Exchange Online module isn't installed, or consent/scopes are missing.

• **Fix:** `Install-Module Microsoft.Graph`, `Install-Module ExchangeOnlineManagement`; sign in with an admin that can consent to the required read scopes (see each script's header). Discovery/Validation/Report are read-only.

A remediation applied the wrong change

• **Cause:** `-Apply` was run without reviewing the `-WhatIf` plan.

• **Fix:** revert using the Phase 1 baseline (accepted-domain type, connector state, DNS). Re-run the script in default `-WhatIf` mode, review, then `-Apply`.

> *After every fix, re-run the corresponding* `05-testing.md` *check and, before declaring resolution, re-run* `GWCOEX-Report.ps1` *to confirm coverage didn't regress elsewhere.*

7. Best Practices — A Stable Transition Period

Coexistence is a **transition state**, not a destination. These practices keep the split organization stable for the migration window and set up a clean exit at consolidation. They apply across all three pillars and reuse the runbook's named objects.

Treat it as one system, piloted and reversible

• **Always pilot first.** Every mail-flow, DNS, and identity change goes through `Coexistence – Pilot Members` before the fleet. Mail routing is the highest-blast-radius pillar — a bad relay or split-delivery rule NDRs real mail.

• **Capture the baseline before you touch anything.** The Phase 1 baseline (MX/SPF/DKIM/DMARC, accepted-domain types, connectors, Interop state) is your rollback. Never change production before it is recorded.

- **Change control on the shared surfaces.** DNS, connectors, the Google routing rule, and GCDS search

rules are the shared surfaces — schedule them in windows with approvals, not ad hoc.

Mail routing / split delivery

- **Lower DNS TTLs before an MX change**, and stage the flip during a low-volume window; raise TTLs

back afterward.

- **Keep exactly one MX-primary** at a time. Both platforms in internal-relay/split mode simultaneously

can create relay loops — validate that a message for a *nonexistent* recipient bounces cleanly rather than ping-ponging.

- **SPF has a 10-DNS-lookup limit** — using `include:_spf.google.com` and

`include:spf.protection.outlook.com` together is fine, but audit the full record so you don't exceed it and softfail legitimate mail.

- **Keep DMARC observable during transition** (`p=none`/`quarantine` with `rua` reporting). Only move to

`p=reject` after Validate/Expand proves both platforms sign and align. Enforcing too early during a two-sender period silently drops mail.

- **Require TLS on both relay hops** (the EXO→Google connector and the Google→M365 host) so relayed mail

stays encrypted in transit.

Free/busy interop

- **Use dedicated role accounts** (`svc-interop-exo@contoso.com`, `svc-interop-gcal@contoso.com`) —

never a real person's account. Document them as service identities; exclude the Exchange one from blocking Conditional Access and (for basic auth) set the password non-expiring.

- **Default to availability-only**, and raise to event-details deliberately, per policy — event-level

privacy is respected on both sides, but titles/locations are still data governance cares about.

- **Prefer the Graph-API endpoint over legacy EWS** where your tenant offers it (modern auth,

no basic-auth password to rotate). If the option is preview/rolling-out, **verify in your tenant**.

- **Protect the endpoint:** the Exchange endpoint must stay reachable on 443 with a valid public-CA

cert; if you restrict inbound, allowlist Google's Interop egress ranges rather than opening it wide.

Directory sync

- **GCDS is one-way and simulate-first.** Always **Simulate sync** and review **Proposed changes**

before a real run — an over-broad search rule can propose mass deletions.

- **Run GCDS on a dedicated host**, not the domain controller; keep it patched and monitored.
- **Represent the other platform's users consistently** — Google **shared contacts** for M365-only recipients, Exchange **mail contacts / mail users** for Google-only recipients — and mark them with `Coexistence – Synced` so they are trivially reportable and removable.
- **One source of truth.** Both platforms provision from the same identity source; never let two systems author the same user or the GAL diverges.

Operate and monitor

- **Weekly coverage report** with `GWCOEX-Report.ps1`; treat any user that loses a pillar as an incident, not a curiosity.
- **Remediation is `-WhatIf` by default** — review the plan before `-Apply`. Same for `GWCOEX-Set-ScopeTargeting.ps1`.
- **Watch NDR and DMARC-report volume** as the leading indicators that a relay or auth change went wrong.

Plan the exit (decommission at consolidation)

Coexistence should end. Build the off-ramp in from day one:

1. **Confirm all mailboxes have migrated** to the surviving platform (data-migration workstream is complete for that user/wave).
1. **Flip MX** fully to the surviving platform and let TTLs settle.
2. **Change the shared domain back to authoritative** on the surviving platform and **disable** `Coexistence – EXO→Google Connector` / the Google `Coexistence – Split Delivery` rule.
1. **Turn off Calendar Interop** endpoints and retire `svc-interop-*` role accounts.
2. **Stop GCDS** and remove the `Coexistence – Synced` contacts/objects on the retired platform.
3. **Tighten DMARC** to a single-sender enforcing policy; remove the retired platform's SPF include and DKIM selector.
1. **Retire the legacy tenant** per the vendor's offboarding process; keep the Phase 1 baseline archived for audit.

> *The measure of a good coexistence design is a **quiet** transition and a **clean** decommission: mail routed, calendars resolved, directory in sync — and nothing left behind when the legacy side is switched off.*

8. Deliverables Summary — Microsoft 365 ↔ Google Workspace Coexistence

Everything in this solution, and what to do with it.

Documents (`source/`)

Doc	Purpose
`01-solution-overview.md`	The coexistence model — the three pillars (mail routing/split delivery, free/busy interop, directory sync), scope, and lifecycle.
`02-requirements.md`	Prerequisites (R1–R9) as Description → Why → Verify → Implement — both platforms + admin, DNS/mail-flow, interop role accounts, directory-sync mechanism, pilot.
`03-implementation.md`	The ordered runbook: **Assess** → **Configure** → **Pilot** → **Validate** → **Expand**, with named objects and per-phase exit gates and rollback.
`05-testing.md`	Pass/fail gates T1–T5 and ongoing regression checks — mail routes, free/busy both ways, directory in sync.
`06-troubleshooting.md`	Symptom → cause → fix for the common coexistence failures (NDRs, relay loops, blank free/busy, missing GAL entries).
`07-best-practices.md`	Keeping the transition stable, and the controlled decommission at consolidation.
`08-deliverables-summary.md`	This file.
`executive-summary.md`	One-page summary for decision-makers.

Scripts (`scripts/`)

All Microsoft-side automation uses the **Microsoft Graph PowerShell SDK** (and the Exchange Online module for mail-flow objects). Discovery, Validation, and Report are **read-only**; Remediation and Set-ScopeTargeting default to `-WhatIf` and require `-Apply` to make changes.

Script	Role	Read/Write
`GWCOEX-Discovery.ps1`	Coexistence-readiness discovery — accepted-domain types, connectors, MX/SPF/DKIM/DMARC, cross-platform contacts, Interop state.	Read-only
`GWCOEX-Validation.ps1`	Per-user interop validation — mail routing, free/busy both directions, directory presence.	Read-only

Script	Role	Read/Write
`GWCOEX-Report.ps1`	Coexistence coverage report — per-user pillar status + overall coverage %.	Read-only
`GWCOEX-Remediation.ps1`	Safe remediation of common gaps (relay type, connector, mail contacts). Default `-WhatIf`.	Write (gated)
`GWCOEX-Set-ScopeTargeting.ps1`	Point the coexistence scope at the pilot group or all users. Default `-WhatIf`.	Write (gated)

How to use it end to end

1. Verify `02-requirements.md` (R1–R9). Capture the DNS + Google + M365 baseline.
2. Run `GWCOEX-Discovery.ps1` for the M365-side baseline.
3. Follow `03-implementation.md` Phase by Phase, honoring each `05-testing.md` gate.
4. In Pilot/Validate, run `GWCOEX-Set-ScopeTargeting.ps1`, `GWCOEX-Validation.ps1`, and `GWCOEX-Report.ps1`; remediate with `GWCOEX-Remediation.ps1` as needed.
5. Expand to the fleet, prove coverage, tighten DMARC.
6. Operate per `07-best-practices.md`; at consolidation, run the **decommission** checklist.

Outcome

Microsoft and Google working as one through the transition — **mail routes, calendars resolve, the directory stays in sync** — piloted, validated, and rollback-ready, until consolidation retires the legacy side cleanly.

Appendix A — PowerShell Scripts

Included in the accompanying ZIP. Each script's header documents full usage.

GWCOEX-Discovery.ps1

Purpose: Build the coexistence-readiness baseline from the Microsoft 365 side and the public DNS of the shared domain. Exports accepted-domain types (authoritative vs internal-relay), inbound/outbound connectors, mail contacts / mail users that represent Google-hosted recipients, and the domain's MX / SPF / DKIM / DMARC. This is the rollback baseline referenced by the runbook. Read-only. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GWCOEX-Discovery.ps1 -SharedDomain your-domain.com -OutputFolder .\discovery (Exchange Online module for mail flow; Microsoft Graph SDK optional for recipients.) Graph scopes if -UseGraph: Organization.Read.All, Directory.Read.All EXPECTED RESULTS Timestamped CSV/TXT files: accepted-domains, connectors, mail-contacts, and a dns-auth summary, plus a console readiness summary. PASS = accepted-domain type, connectors, and DNS auth are all captured for the shared domain. No changes made. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #>

How to run: powershell -ExecutionPolicy Bypass -File .\GWCOEX-Discovery.ps1 -SharedDomain your-domain.com -OutputFolder .\discovery (Exchange Online module for mail flow; Microsoft Graph SDK optional for recipients.) Graph scopes if -UseGraph: Organization.Read.All, Directory.Read.All EXPECTED RESULTS Timestamped CSV/TXT files: accepted-domains, connectors, mail-contacts, and a dns-auth summary, plus a console readiness summary. PASS = accepted-domain type, connectors, and DNS auth are all captured for the shared domain. No changes made. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #>

GWCOEX-Remediation.ps1

Purpose: Remediate the most common Microsoft-side coexistence gaps: set the shared domain to Internal relay, create the outbound relay connector to Google's smart host, and create mail contacts for Google-hosted recipients that are missing from the GAL. Every change is gated: the script runs in preview (-WhatIf) mode by default and only makes changes when you pass -Apply. It always prints the plan first. HOW TO RUN Preview (default): powershell -ExecutionPolicy Bypass -File .\GWCOEX-Remediation.ps1 -SharedDomain your-domain.com Apply after reviewing the plan: powershell -ExecutionPolicy Bypass -File .\GWCOEX-Remediation.ps1 -SharedDomain your-domain.com -Apply (Optional -MissingContactsCsv columns: DisplayName,ExternalEmailAddress) Uses the Exchange Online module; run as an Exchange administrator. EXPECTED RESULTS In preview: a numbered list of the changes that WOULD be made, no tenant change. With -Apply: those changes are made and confirmed. PASS = shared domain is Internal relay and an enabled outbound route to Google exists. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #>

How to run: Preview (default): powershell -ExecutionPolicy Bypass -File .\GWCOEX-Remediation.ps1 -SharedDomain your-domain.com Apply after reviewing the plan: powershell -ExecutionPolicy Bypass -File .\GWCOEX-Remediation.ps1 -SharedDomain your-domain.com -Apply (Optional -MissingContactsCsv columns: DisplayName,ExternalEmailAddress) Uses the Exchange Online module; run as an Exchange administrator. EXPECTED RESULTS In preview: a numbered list of the changes that WOULD be made, no tenant change. With -Apply: those changes are made and confirmed. PASS = shared domain is Internal relay and an enabled outbound route to Google exists. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #>

GWCOEX-Report.ps1

Purpose: Roll up per-user interop validation into a coexistence coverage report. Reads a validation CSV produced by GWCOEX-Validation.ps1 (or re-derives per-pillar status) and produces a single coverage view: how many users are green on mail routing, free/busy, and directory, plus an overall coverage percentage and a per-user detail table. Read-only reporting — makes no changes. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GWCOEX-Report.ps1 -ValidationCsv .\validation\validation-<stamp>.csv (No Graph scopes required when reading an existing validation CSV.) EXPECTED RESULTS A console coverage summary and an HTML + CSV report in -OutputFolder. PASS = the overall coverage percentage meets or exceeds -Threshold (default 100). No changes. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #>

How to run: powershell -ExecutionPolicy Bypass -File .\GWCOEX-Report.ps1 -ValidationCsv .\validation\validation-<stamp>.csv (No Graph scopes required when reading an existing validation CSV.) EXPECTED RESULTS A console coverage summary and an HTML + CSV report in -OutputFolder. PASS = the overall coverage percentage meets or exceeds -Threshold (default 100). No changes. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #>

GWCOEX-Set-ScopeTargeting.ps1

Purpose: Control how widely coexistence mail routing applies. During Pilot, the outbound relay to Google is scoped to the pilot group via a transport rule so only pilot recipients relay; during Expand, the scope is widened to all users. This script creates or updates the scoping transport rule for the coexistence connector. It runs in preview (-WhatIf) mode by default and only changes the tenant with -Apply. HOW TO RUN Preview scoping to the pilot group (default): powershell -ExecutionPolicy Bypass -File .\GWCOEX-Set-ScopeTargeting.ps1 -Scope Pilot -PilotGroup 'Coexistence - Pilot Members' Apply expand to all users: powershell -ExecutionPolicy Bypass -File .\GWCOEX-Set-ScopeTargeting.ps1 -Scope AllUsers -Apply Uses the Exchange Online module; run as an Exchange administrator. EXPECTED RESULTS In preview: the scope change that WOULD be made (pilot-group-only vs all users). With -Apply: the transport rule is created/updated accordingly. PASS = the active scope matches the requested -Scope. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #>

How to run: Preview scoping to the pilot group (default): powershell -ExecutionPolicy Bypass -File .\GWCOEX-Set-ScopeTargeting.ps1 -Scope Pilot -PilotGroup 'Coexistence - Pilot Members' Apply expand to all users: powershell -ExecutionPolicy Bypass -File .\GWCOEX-Set-ScopeTargeting.ps1 -Scope AllUsers -Apply Uses the Exchange Online module; run as an Exchange administrator. EXPECTED RESULTS In preview: the scope change that WOULD be made (pilot-group-only vs all users). With -Apply: the transport rule is created/updated accordingly. PASS = the active scope matches the requested -Scope. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #>

GWCOEX-Validation.ps1

Purpose: For each user in a supplied list, validate the three coexistence pillars from the Microsoft 365 side: mail routing (the shared domain relays and a route to Google exists), free/busy readiness (the Calendar Interop role accounts and an Exchange availability endpoint are in place), and directory presence (a Google- hosted counterpart is represented as a mail contact / mail user in the GAL). Read-only — it inspects configuration and object state, it does not send mail. HOW TO RUN powershell -ExecutionPolicy Bypass -File .\GWCOEX-Validation.ps1 -SharedDomain your-domain.com -UserListCsv .\pilot-users.csv (pilot-users.csv columns: UserPrincipalName,Platform where Platform = M365 or Google) Graph scopes if -UseGraph: Directory.Read.All EXPECTED RESULTS A per-user results table and a

CSV: MailRoute, FreeBusy, Directory each PASS/FAIL. PASS for a user = all three pillars OK. No changes are made. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #>

How to run: powershell -ExecutionPolicy Bypass -File .\GWCOEX-Validation.ps1 -SharedDomain your-domain.com -UserListCsv .\pilot-users.csv (pilot-users.csv columns: UserPrincipalName,Platform where Platform = M365 or Google) Graph scopes if -UseGraph: Directory.Read.All EXPECTED RESULTS A per-user results table and a CSV: MailRoute, FreeBusy, Directory each PASS/FAIL. PASS for a user = all three pillars OK. No changes are made. .NOTES White-label — the repeatable way. Read-only against Microsoft Graph. #>